



QUALITY MANAGEMENT BRANCH

STANDARD OPERATING PROCEDURES

FOR

VERIFICATION AND CALIBRATION OF TEMPERATURE DEVICES

Standards Laboratory SOP 006

First Edition

MONITORING AND LABORATORY DIVISION

JULY 2014



California Environmental Protection Agency

AIR RESOURCES BOARD

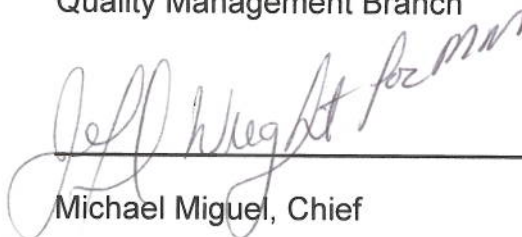
Standard Operating Procedures (SOP) Approval

Title: Verification and Calibration of Temperature Devices
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Branch: Quality Management Branch
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Prepared by: Robert Russell, Air Pollution Specialist
Approval: This SOP has been reviewed and approved by:


Jeff Wright, Manager

Data Analysis and Special Projects Section

Quality Management Branch


Michael Miguel, Chief

Quality Management Branch



DATE



DATE

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1.0 Introduction

This procedure will provide National Institute of Standards and Technology (NIST) traceable verifications and calibrations for temperature devices in the range of 0.00 to 50.00 °C. Platinum Resistance Thermometers (PRT), Thermistors, and Thermocouple temperature sensors are acceptable. Probe dimension limitation is 1/8" to 1/2" diameter. All hand-held digital temperature meters are generally acceptable.

2.0 Regulatory Requirements

U.S. Environmental Protection Agency guidance document: Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements, Version 2.0, describes the types of temperature measuring devices in the field environment, including the operation and maintenance, calibration, and auditing procedures. The recommended tolerance for a calibration or audit is 0.5 °C.

3.0 Summary of Method

Temperature probe devices are compared against a NIST traceable PRT at five points: 0, 20, 30, 40, and 50 °C. If the comparison meets the verification criteria (± 0.5 °C at each point), a verification report is issued. No adjustments or corrections are needed before using the data from the display of the candidate device. If an adequate coefficient of determination (R^2) is developed from comparing the candidate device to the reference standard ($R^2 \geq 0.999$), a calibration correction equation is also provided. A client may choose to use this calibration formula to correct the candidate's display to improve the accuracy of the readings.

If the temperature device does not pass, a re-test is performed to confirm the failure. If the re-test confirms the failed criteria, the device must be repaired by the client before further testing is performed.

4.0 Summary of Changes to Previous Version

Not Applicable

5.0 Definitions

- Calibration – establishes a correction formula to be used to adjust or correct the display of the candidate instrument. This is determined through a comparison of the candidate instrument to a

known, or reference standard.

- Verification – establishes comparability of a candidate instrument to a known, or reference standard; the output of candidate instrument is not corrected based upon the results of the verification procedure.

6.0 Personnel Qualifications

Before new personnel perform this procedure, one or more weeks of training from Standards Laboratory staff is required. Upon completion of this training, new personnel must be able to demonstrate competency in performing this procedure without assistance.

7.0 Health and Safety

Because of the design of temperature probes, care should be taken when handling the probes to avoid injury.

8.0 Cautions

- **DO NOT** use pliers or other similar devices to squeeze the sheathing of the temperature sensor around the probe wiring harness. This could permanently damage the sensor and change the calibration coefficient.
- **DO NOT** use fluid to clean inside the well of the calibrator.
- It is important to keep the well of the calibrator clean and clear of any foreign matter.

9.0 Interferences

- Avoid operating the instrument in an oily, wet, or dirty environment.
- Operating the instrument in a draft-free environment improves performance of the instrument.

10.0 Equipment and Supplies

- Platinum Resistance Thermometer (PRT).

- Model 9142 Fluke Field Metrology Well (-25 to 150 °C).

11.0 Verification/Calibration Procedure

11.1 Power up the Fluke Metrology Well. See Figure 1.



Figure 1: Fluke Metrology Well Power Switch

11.2 The metrology well has six hole sizes. Insert the Fluke reference PRT into the hole that best matches the diameter of the probe. Insert the probe until it bottoms out; then raise it about one inch. See Figure 2.

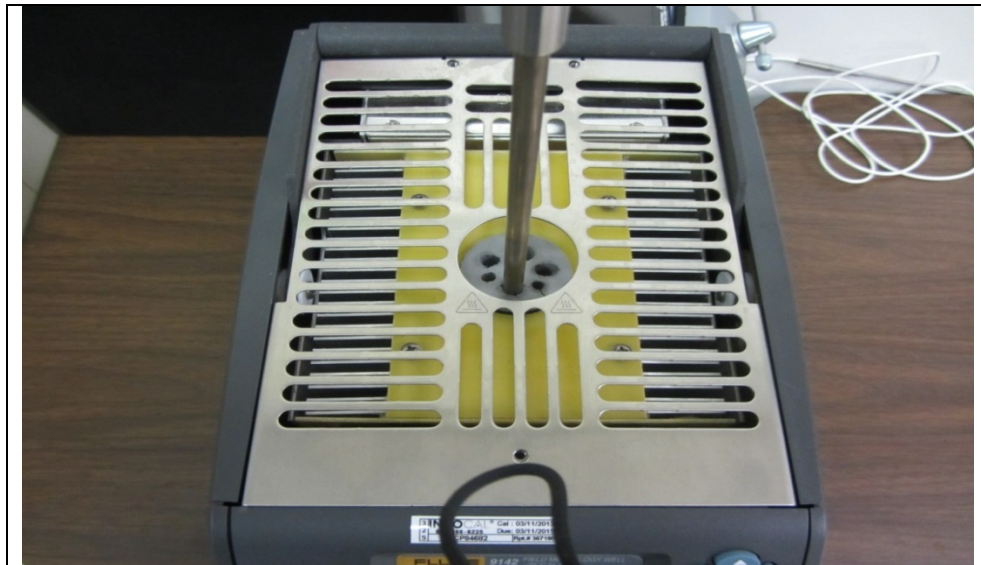


Figure 2: Reference Probe Inserted

11.3 Insert the candidate temperature probe into the hole that best matches the diameter of the probe. See Figure 3. Do not force the probe into the well.

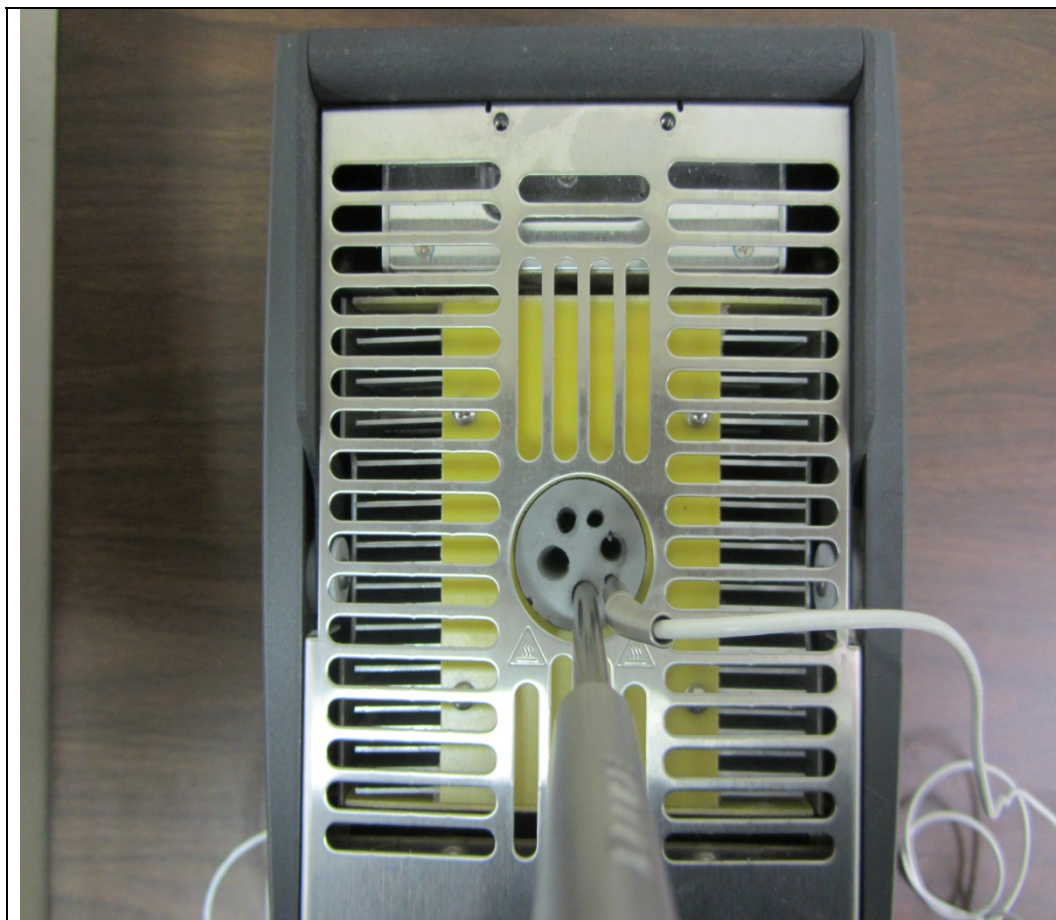


Figure 3: Client's Temp. Probe Inserted

- 11.4 Power up the candidate temperature standard. Allow a 15 minute warm-up before starting the test.
- 11.5 Locate the HP laptop computer on the table across from the Fluke reference temperature standard. Double-click on the "CalReport" icon. See Figure 4.

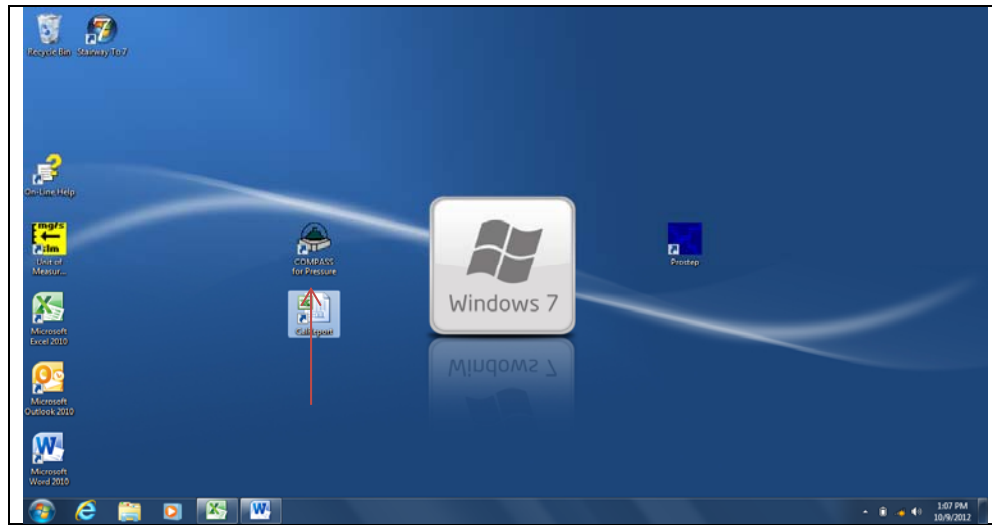


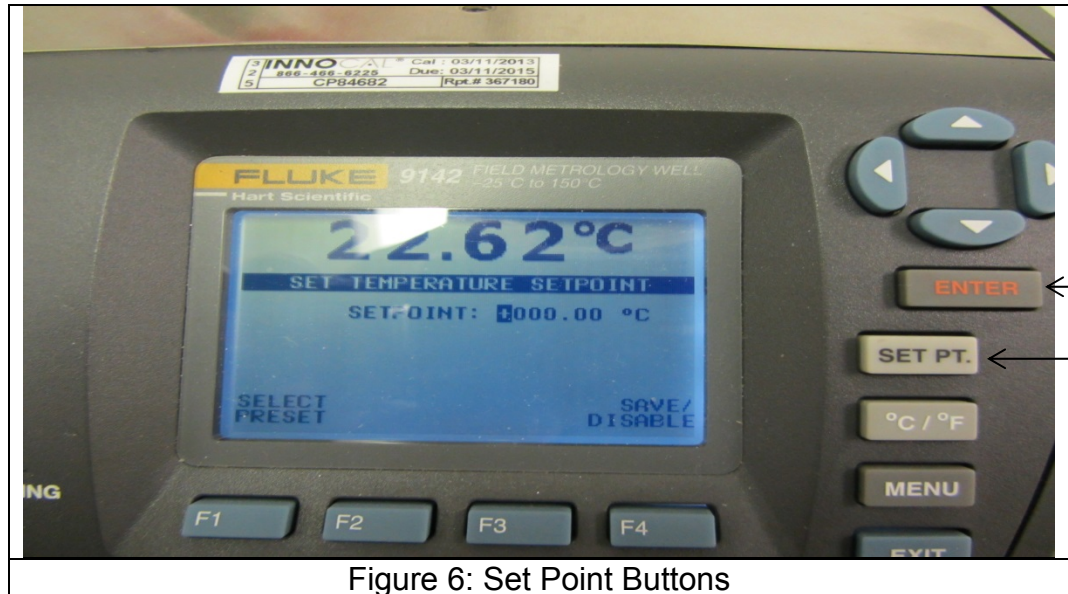
Figure 4: "CalReport" Workbook Icon

- 11.6 Click on the "TemperatureProbeTest" spreadsheet. Fill in the blue shaded cells with the following information: Room Temperature, Room Pressure, and %RH. This information is located on the Vaisala all-in-one meter located at the top of the gas analyzer instrument rack. Enter the Log number, Customer, Description, Barcode, Serial #, Manufacturer, and Calibration Date. See Figure 5.

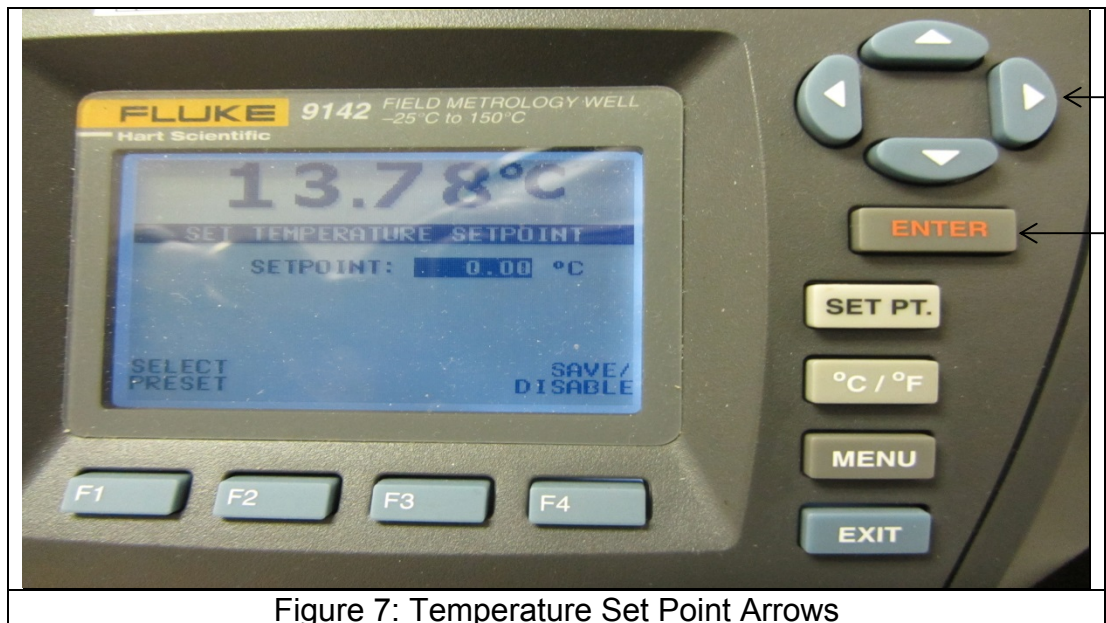
CalReport [Compatibility Mode] - Microsoft Excel					
File Home Insert Page Layout Formulas Data Review View					
Clipboard Font Paragraph Styles Cells Editing					
A18 X % RH					
5	A	B	C	D	E
6					
7					
8					Log # 2014043
9					Device Under Test
10	Temperature Reference Standard			Customer:	BAAQMD
11	Test Facility:	ARB/MLD/STDLAB		Description:	RH/TEMP
12	Description:	Metrology Well		Barcode	106058
13	Model number	9142		Serial #:	D4420059
14	Serial Number	20071242		Manufacturer:	VAISALA
15				Calibration Date:	2/7/2014
16	Room Temperature °C:	24.1			
17	Room Pressure mmHg:	764.4			
18	% RH	37.2			
19					
TEMPROBTEST DeltaCal TetraCal Pressure Std. Verif. Barometer Report Manometer Report Rotometer Chinook Temp. Ref. Std. Verification RH Report					

Figure 5: "TemperatureProbeTest" Spreadsheet

- 11.7 Press the “SET PT.” button on the front panel of the metrology well; then press the enter button. See Figure 6.



- 11.8 Use the arrow buttons to set the temperature to 0.00 °C; then press the “ENTER” button 2 times. The metrology well will cool to 0.00 °C. There will be an audible beep when the temperature set point has been reached. See Figure 7.



- 11.9 Record the “Ref. Probe” temperature reading and the “Candidate Probe” temperature reading onto the spreadsheet. See Figure 8. For the location of reference temperature reading, see Figure 9.

SET POINT	Ref. Probe °C	Candidate Probe °C	Difference °C	PASS/FAIL
0.0	-0.23	-0.01	0.22	PASS
20.0	19.73	20.02	0.29	PASS
30.0	29.72	30.02	0.30	PASS
40.0	39.72	40.04	0.32	PASS
50.0	49.73	50.06	0.33	PASS

SLOPE	INTERCEPT	R2	R2 ≥ 0.999
1.00218	0.23094	0.9999999	Pass

Temperature Reading Uncertainty of the Reference Probe is +/- 0.21 deg. C Full Range

Figure 8: Reference and Candidate Probe Temperature Readings



Figure 9: Reference Temperature Reading

- 11.10 Repeat steps 11.7 through 11.09 at 20.00 °C Set Point.
- 11.11 Repeat steps 11.7 through 11.09 at 30.00 °C Set Point.
- 11.12 Repeat steps 11.7 through 11.09 at 40.00 °C Set Point.
- 11.13 Repeat steps 11.7 through 11.09 at 50.00 °C Set Point.
- 11.14 Press the “SET PT.” button then press “F4” to save and end the test.
- 11.15 Turn off power to the Fluke Metrology Well and the candidate temperature device.
- 11.16 Carefully remove the temperature probes one at a time from the well.
- 11.17 If the candidate temperature reading differs by more than $\pm 0.5^{\circ}\text{C}$ at any point as indicated in the Pass/Fail column (Figure 10), perform a re-test to confirm the failure. If confirmed, inform the client that the temperature device did not meet criteria and proceed with a SFN (Standard Failure Notification). The SFN is found in the following location: W:\Standards Laboratory\Working Tables. Otherwise, make sure the spreadsheet is accurate and complete.

Log #	2014043			
Device Under Test				
Customer:	BAAQMD			
Description:	RH/TEMP			
Barcode	106058			
Serial #:	D4420059			
Manufacturer:	VAISALA			
Calibration Date:	2/7/2014			
				↓
SET POINT	Ref. Probe °C	Candidate Probe °C	Difference °C	PASS/FAIL
0.0	-0.23	-0.01	0.22	PASS
20.0	19.73	20.02	0.29	PASS
30.0	29.72	30.02	0.30	PASS
40.0	39.72	40.04	0.32	PASS
50.0	49.73	50.06	0.33	PASS
SLOPE	INTERCEPT	R2	R2 ≥ 0.999	
1.00218	0.23094	0.9999999	Pass	
Temperature Reading Uncertainty of the Reference Probe is +/- 0.21 deg. C Full Range				

Figure 10: Temperature Probe Test Spreadsheet

11.18 Print one copy of the spreadsheet and save and close the “CalReport” workbook. Place this copy inside the client’s folder.

12.0 Calibration Report Generation

12.1 Double click on the “DASPS” icon. You can find the DASPS icon on any computer located in the Standards Laboratory. See Figure 11.



12.2 Double click on the “Standards Laboratory” folder. See Figure 12.

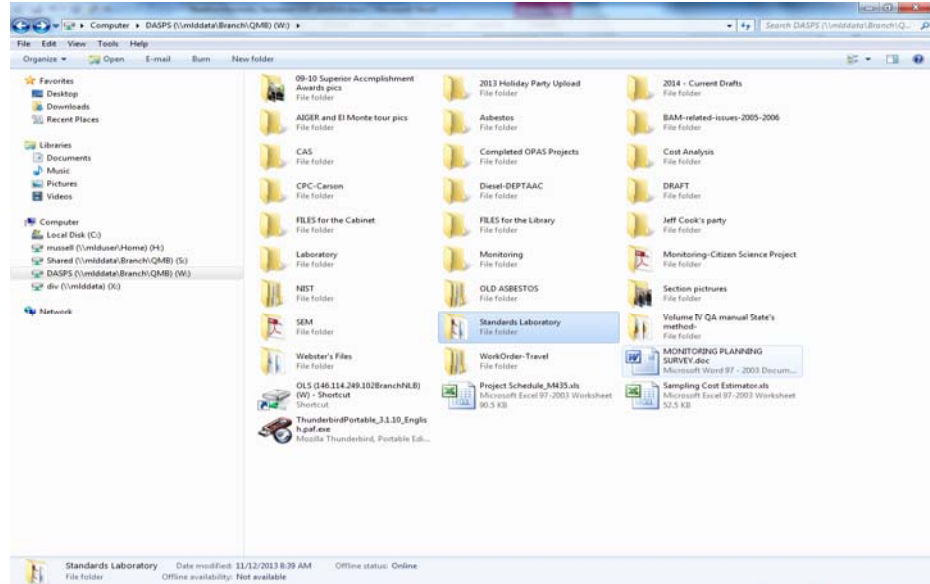


Figure 12: “Standards Laboratory”

12.3 Double click on the “Calibration Forms” folder. See Figure 13.

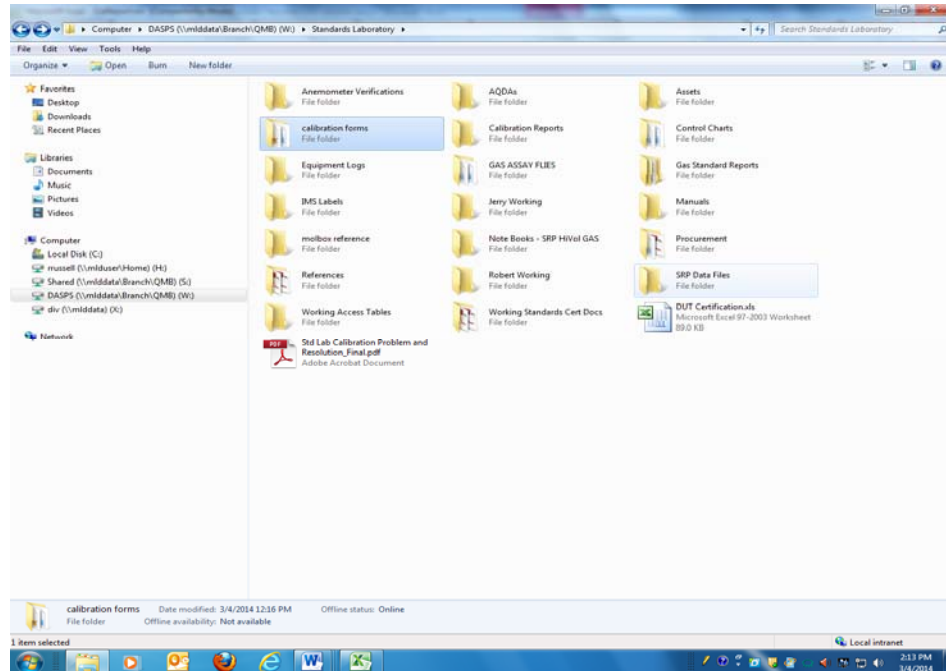
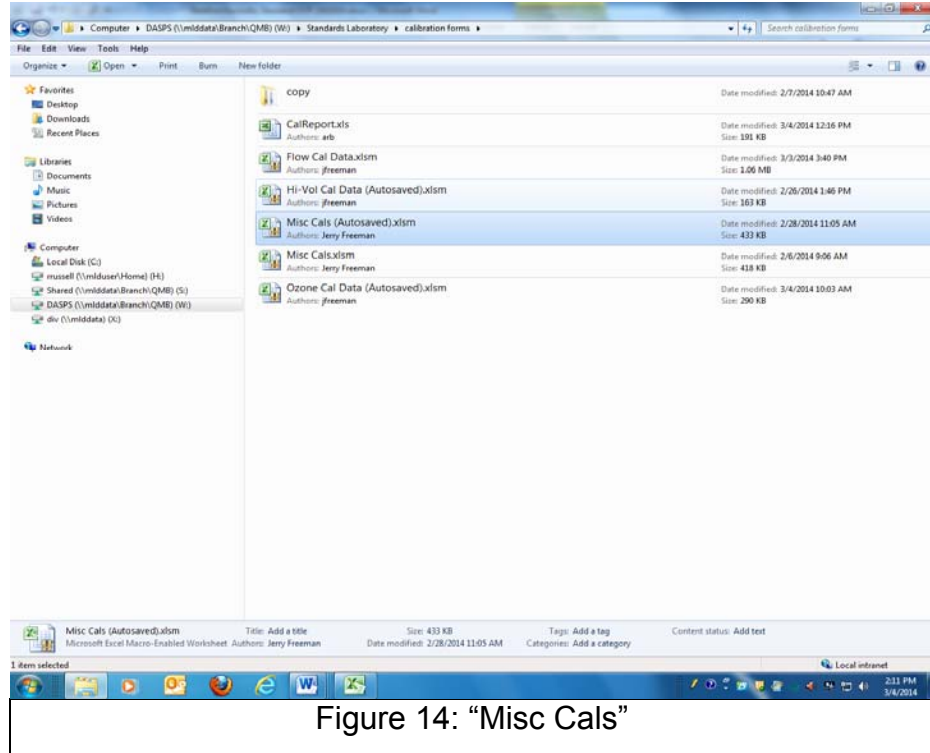
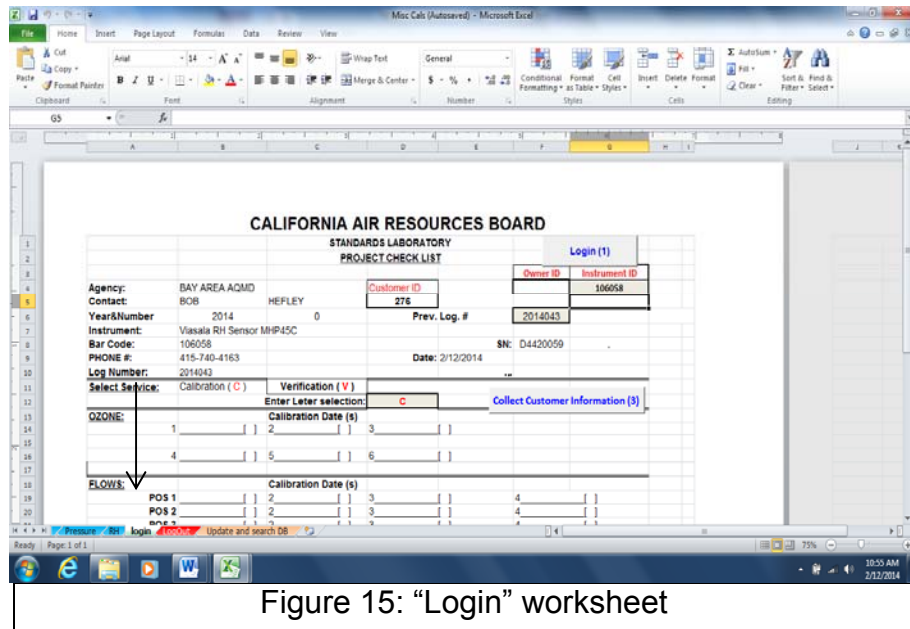


Figure 13: “Calibration Forms”

12.4 Double click on the “Misc Cals” folder. See Figure 14.



12.5 Click on the “Login” worksheet. See Figure 15.



12.6 Enter instrument ID. See Figure 16.

CALIFORNIA AIR RESOURCES BOARD
STANDARDS LABORATORY
PROJECT CHECK LIST

Agency: BAY AREA AQMD
Contact: BOB HEFLEY
Year&Number: 2014 0
Instrument: Viasala RH Sensor MHP45C
Bar Code: 106058
PHONE #: 415-740-4163
Log Number: 2014043
Date: 2/12/2014
SN: D4420059

Select Service: Calibration (C) Verification (V)
Enter Letter selection: C
Collect Customer Information (3)

OZONE: Calibration Date (s)
1 2 3 4 5 6

FLWS: Calibration Date (s)
POS 1 2 3 4
POS 2 2 3 4
POS 3 2 3 4
POS 4 2 3 4

HIVOLL OFFICE
Calibration Date (s) 1 2 3

TEMPERATURE
Calibration Date (s) 1 2 3

Figure 16: Instrument barcode

12.7 Click on the “Collect Customer Information” button. See Figure 17.

CALIFORNIA AIR RESOURCES BOARD
STANDARDS LABORATORY
PROJECT CHECK LIST

Agency: BAY AREA AQMD
Contact: BOB HEFLEY
Year&Number: 2014 0
Instrument: Viasala RH Sensor MHP45C
Bar Code: 106058
PHONE #: 415-740-4163
Log Number: 2014043
Date: 2/21/2014
SN: D4420059

Select Service: Calibration (C) Verification (V)
Enter Letter selection: C
Collect Customer Information (3)

OZONE: Calibration Date (s)
1 2 3 4 5 6

FLWS: Calibration Date (s)
POS 1 2 3 4
POS 2 2 3 4
POS 3 2 3 4
POS 4 2 3 4

HIVOLL OFFICE
Calibration Date (s) 1 2 3

TEMPERATURE
Calibration Date (s) 1 2 3

Figure 17: “Collect Customer information”

12.8 Click on the “Temp” worksheet. See Figure 18.

TEMPERATURE TEST REPORT

To: BAY AREA AQMD
BOB HEFLEY

Log Number: 2014043

From: Robert Russell, Jerry Freeman
Data Analysis and Special Projects Section

Calibration Date: 1/9/2014
Report Date: 2/21/2014

IDENTIFICATION

Instrument: Viasala RH Sensor MHP45C
Property No.: 106058
Serial No.: D4420059
Previous Log No.: N/A
Bar Code No.: 106058
Elevation: 25.00
Property of: BAY AREA AQMD

Site Name: CARB
Location: 1927 13th Street
Sacramento, CA

Customer Number: 276

CALIBRATION STANDARDS

Calibration Standards	Expiration Date	ID Number
FLUKE 9143 METROLOGY WELL	20071242	20071242

LINEAR RELATIONSHIP

Figure 18: “Temp” worksheet

12.9 Transfer the temperature readings in the “Ref Probe” column (Figure 10) to the “Reference Output” column (Figure 19). Transfer the temperature readings in the “Candidate Probe” column (Figure 10) to the “Candidate Output” column (Figure 19).

The linear relationship table is for informational use only. DO NOT USE THESE VALUES TO CORRECT THE INSTRUMENT DISPLAY.

Reference Output Temperature	Candidate Output Temperature	Difference	Pass/Fail
-0.23	-0.01	0.221	PASS
19.73	20.02	0.286	PASS
29.72	30.02	0.300	PASS
39.72	40.04	0.320	PASS
49.73	50.06	0.330	PASS

Expiration Date: 1/9/2015

CALIBRATION EQUATION:

Corrected Temperature = 0.990 * (Net Display) + 0.230

The Correction Equation can be used to correct the instrument display to improve the accuracy of the reading.

Figure 19: “Temperature Test Report” worksheet

12.10 Click on the “RUN” button once to save data. See Figure 20.

The screenshot shows a Microsoft Excel spreadsheet titled "Misc Cals (Autosaved) - Microsoft Excel". The spreadsheet is a "TEMPERATURE TEST REPORT" form. The form is organized into several sections:

- Header:** "Run" button in cell D34.
- General Information:**
 - To: BAY AREA AQMD
 - Log Number: 2014043
 - From: Robert Russell/ Jerry Freeman
 - Calibration Date: 1/9/2014
 - Data Analysis and Special Projects Section
 - Report Date: 2/21/2014
- IDENTIFICATION:**
 - Instrument: Viasala RH Sensor MHP45C
 - Property No.: 106058
 - Serial No.: D4420059
 - Previous Log No.: N/A
 - Bar Code No.: 106058
 - Elevation: 25.00'
 - Property of: BAY AREA AQMD
 - Customer Number: 276
- CALIBRATION STANDARDS:**
 - FLUKE 9143 METROLOGY WELL
 - Expiration Date: (blank)
 - ID Number: 20071242
- LINEAR RELATIONSHIP:** (Section header, no data entered)

The spreadsheet is displayed in the "Formulas" tab. The status bar at the bottom shows "Ready" and the system clock is 2:56 PM on 2/21/2014.

Figure 20: “Temperature Test Report”

12.11 Have a peer review the “Temperature Test Report” for completeness and accuracy. Print two copies of the test report. Place one copy with the client’s temperature device and file the other copy along with the “CalReport” spreadsheet inside the customer’s login folder. Convert the Temperature Test Report to a PDF document. Save in the following location: W:\Standards Laboratory\Calibration Reports.

13.0 Quality Control

- A semi-annual multipoint comparison is performed between the Fluke temperature reference standard and the Omega HH41 Digital Thermometer. The acceptance criteria are ± 0.5 °C at each point and an $R^2 \geq 0.9999$. If all criteria are not met, re-run the test to confirm the result. If confirmed, consult with lead staff to determine if the problem is with the control standard or reference standard. If it is determined to be an issue with the control standard or the reference standard, have the device sent to a qualified vendor that utilizes NIST traceable standards for repair or re-calibration. If the test meets all criteria, the slope and intercept is added to the temperature control chart located in the following location:
W:\Standards Laboratory\Control Charts.

14.0 Troubleshooting

- If the reference or candidate temperature device does not reach the set point within ± 0.5 °C, check to see if the probe was inserted according to section 11.2.
- If the temperature well does not operate according to this operation procedure, refer to the Troubleshooting Guide on page 101 of the "Field Metrology Well Technical Guide."

15.0 References

- Model 9142 Fluke Field Metrology Well Users Guide.
- U.S. EPA Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements. Version 2.0.